

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 03: Probability Distributions: Discrete and Continuous distributions

LECTURE W3L1: Discrete Probability Distributions

BY

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CONCEPTS COVERED

Discrete Probability Distributions

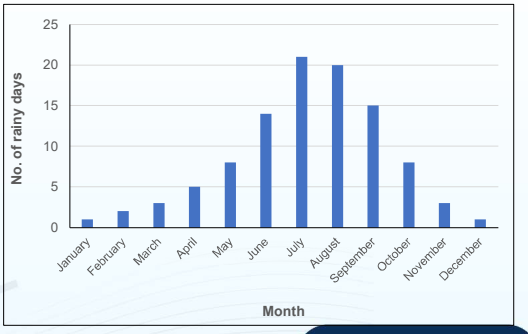
- **Binomial Distribution**
- **Negative Binomial Distribution**
- **Hypergeometric Distribution**
- **Geometric Distribution**
- **Poisson Distribution**
- **Multinomial Distribution**

Discrete Random Variable

- A discrete random variable can take only finite or countably infinite distinct values.
- These values are usually obtained by counting.
- Mathematically, if X is a discrete random variable, then it may take values such as:

$$X = 0, 1, 2, 3, \ldots$$

Examples - number of rainy days in a month, number of occurrences of an extreme event in a year, etc.



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Discrete Probability Distribution

- A discrete probability distribution describes the probability of occurrence of each possible value of a discrete random variable.
- For example, let X be a random variable representing the number of rainy days in a month at a location.
- Since X can take only non-negative integer values, it represents a discrete random variable.
- In discrete probability distributions, probabilities are represented using a Probability Mass Function (PMF).
- Discrete probability distributions are widely used in hydrology and civil engineering for modelling count-based events such as rainy days, embankment failures, etc.

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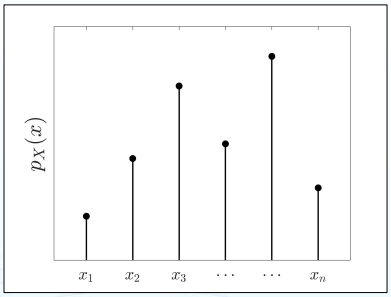
Probability Mass Function (PMF)

- Let us consider X to be a discrete random variable taking values in a set $\theta = \{x_1, x_2, \dots, x_n\}$. The probability mass function (PMF) of X is $\mathrm{p}_X(\bullet)$ satisfying,

  (i) $\mathrm{p}_X(x_i) \geq 0 \; \forall x_i \in \theta$

  (ii) $\sum_{all\ i} \mathrm{p}_X(x_i) = 1$

- A typical plot of a PMF is shown in Fig, where filled circles indicates the probability masses concentrated at a point.
- The vertical lines as such, does not indicate anything except showing the position of the values on the x-axis.



Cumulative Distribution Function (CDF)

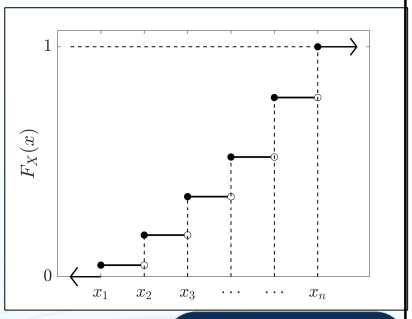
- The CDF ($F_X(x_i)$) represents the probability that X is less than or equal to x_i. This can be represented as

$$F_X(x_i) = P(X \leq x_i) = \sum_{j=1}^{i} P(X = x_j)$$

$$\forall\, x_i \in \{x_1, x_2, \dots, x_n\}$$

- A typical plot of CDF for the discrete random variable is shown in Fig. which is a non-decreasing, discontinuous, staircase-like functions with irregular rise.
- The probability that $X = x_i$ can be determined from the CDF as follows:

$$p_X(x_i) = F_X(x_i) - F_X(x_{i-1})$$



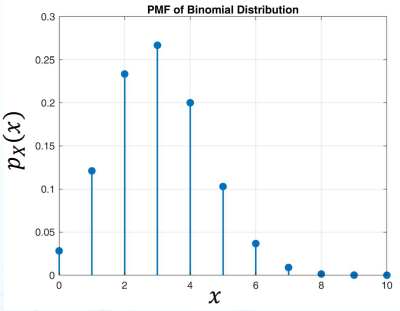
Binomial Distribution

- It is used when an experiment consists of a fixed number of independent trials (n).
- Each trial has only two possible outcomes: Occurrence (p) and Non-occurrence $(1 - p = q)$.
- The probability of occurrence remains the same in every trial.
- If X represents the number of occurrences in n trials, then X follows a binomial distribution.
- The Probability Mass Function (PMF) is given by:

$$p_X(x; n, p) = {}^{n}C_r \, p^x \, (1 - p)^{n-x}, \qquad x = 0,1,2, \ldots, n$$

where, x = no. of occurrences.

- The outcome of any trial is independent of the history of any prior occurrence.



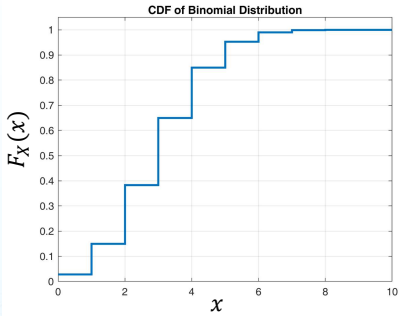
Binomial Distribution

- The CDF is expressed as:

$$F_X(x; n, p) = \sum_{i=0}^{x} p_X(i; n, p)$$

- The mean, variance and coefficient of skewness of the binomial distribution are :

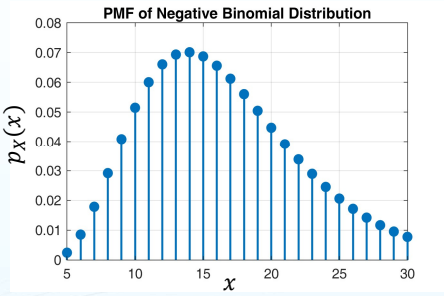
$$E(X) = np; \qquad Var(X) = npq; \; \gamma = \frac{(q - p)}{\sqrt{npq}}$$



Applications:

- Used to estimate the chance that an extreme event will occur or exceed a fixed threshold (probability of exceedance).
- Used to assess the probability that an extreme design event may be equaled or exceeded during the design life of a hydraulic structure, even when a high return period is selected.

Negative Binomial Distribution

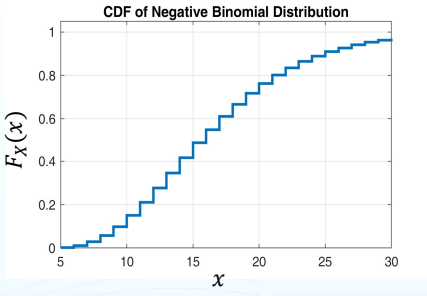


- Negative binomial distribution is a discrete probability distribution used for a sequence of independent Bernoulli trials.
- The number of occurrence (X) in a sequence of independent and identically distributed Bernoulli trials before a specific number of non-occurrence occurs.
- The Probability Mass Function (PMF) is given by:

$$p_X(x; \mathrm{j}\, p) = {}^{x-1}C_{j-1}\, p^{\mathrm{j}}\, (1-p)^{x-\mathrm{j}}, \qquad x = \mathrm{j}, \mathrm{j}+1, \ldots$$

where, j is the number of non-occurrences and p is the probability of occurrence in each independent trial.

Negative Binomial Distribution



- The CDF is expressed as:

$$F_X(x; j, p) = \sum_{i=j}^{x} p_X(i; j, p)$$

- The mean, variance and coefficient of skewness of the negative binomial distribution are :

$$E(X) = \frac{j}{p} \qquad Var(X) = \frac{j(1-p)}{p^2}$$

$$\gamma = \frac{(1+p)}{\sqrt{pj}} \quad \text{where, } q = (1-p).$$

Applications:

- Useful for modelling rare events such as thunderstorms, hail days, tropical cyclones, and hurricanes over a given period (Rare-event frequency).

Hypergeometric Distribution

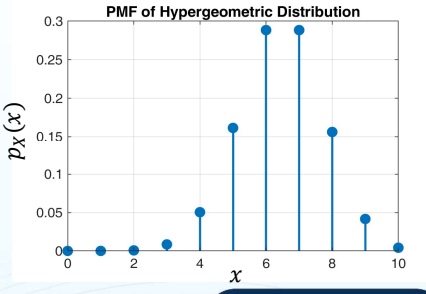
- It is used when the number (X) of occurrences of an event in a sample of size n (drawn without replacement), from a population of size N containing k specific possibilities of occurrences.
- The Probability Mass Function (PMF) is given by:

$$p_X(x; N, n, k) = \frac{{}^{k}C_x \times {}^{N-k}C_{n-x}}{{}^{N}C_n},$$

$$\text{for} \quad x = \max(0, n + k - N), \ldots, \min(n, k)$$

- The CDF is expressed as:

$$F_X(x; N, n, k) = \sum_{x=\max(0, n+k-N)}^{x} p_X(i; n, p)$$



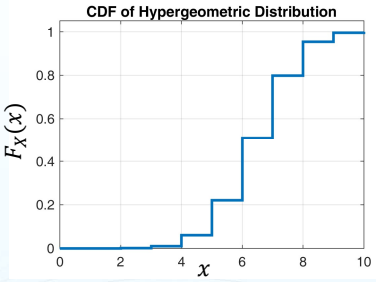
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Hypergeometric Distribution

- The mean, variance and coefficient of skewness of the hypergeometric distribution are :

$$E(X) = \frac{nk}{N} \qquad Var(X) = \frac{nk(N-k)(N-n)}{N^2(N-1)}$$

$$\gamma = \frac{(N-2k)(N-2n)(N-1)^{1/2}}{\sqrt{nk(N-k)(N-n)}\ (N-2)}$$

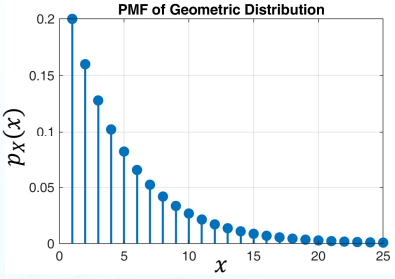


Applications:

- Used to test the significance of relationships between climate indices and hydrologic variables.
- Used to find the probability of favorable cases (k) in the sample (n) from a known population (N).

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Geometric Distribution



- The probability that the first success of a Bernoulli trial occurs on the xth trial can be found using the geometric distribution.
- Here, the random variable is the number of trials (X) in the sequence of a Bernoulli process to get the first occurrence.
- The Probability Mass Function (PMF) is given by:

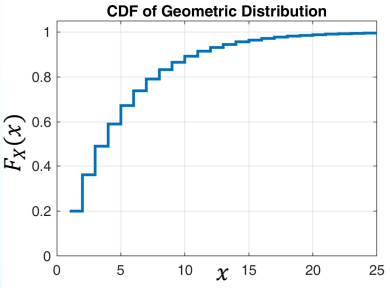
$$p_X\,(x;p) = p(1-p)^{x-1} \quad for\ x = 1,2,\dots,n$$

- The CDF is expressed as:

$$F_X(x;p) = \sum_{i=1}^{x} p_X(i;p)$$

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Geometric Distribution



- The mean, variance and coefficient of skewness of the geometric distribution are :

$$E\,(X) = \frac{1}{p}; \quad Var(X) = \frac{(1-p)}{p^2}$$

$$\gamma = \frac{(2-p)}{\sqrt{(1-p)}}$$

Applications:

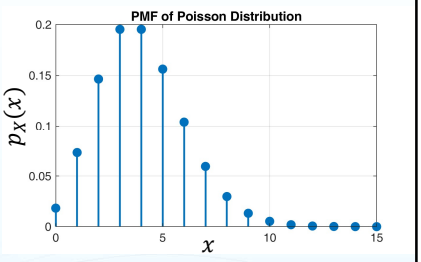
- Used to estimate the probability that a hydrologic event, such as embankment overtopping, cyclone, or extreme rainfall, occurs for the first time after a sequence of non-occurrences.

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Poisson Distribution

- It is used to model the number of events occurring within a fixed interval of time, space, or area.
- Here, the random variable is: the number of occurrences (X) of an event (outcomes of a Bernoulli process) in a fixed interval of time.
- The Probability Mass Function (PMF) is given by:

$$p_X(x;\lambda) = \lambda^x \frac{e^{-\lambda}}{x!}, for\ x = 0,1,\ldots;\lambda > 0$$



- A process is defined as a Poisson process if the events occurring over time/area/space satisfy the three assumptions:
  a) The number of events occurring in disjoint time intervals is independent.

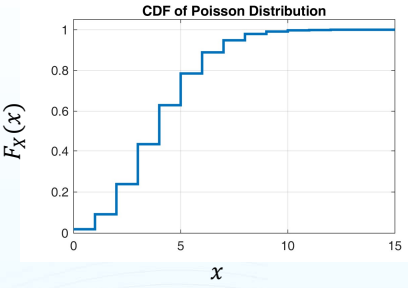
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Poisson Distribution

  b) The probability of a single occurrence in a small time interval is proportional to the length of the interval.
  c) Probability of more than one occurrence in a small interval is negligible.

- The CDF is expressed as:

$$F_X(x;p) = \sum_{i=0}^{x} p_X(i;p)$$



- The mean, variance and coefficient of skewness of the Poisson distribution are :

$$E(X) = \lambda \qquad Var(X) = \lambda \qquad \gamma = \lambda^{-1/2}$$

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Poisson Distribution

Applications:

- Used to estimate the number of occurrences of rare events such as thunderstorms, hail days, cyclones, hurricanes, or floods within a fixed time period.
- Used to estimate the number of rainy days in a particular monsoon month.
- Useful for comparing event occurrence patterns over different time periods, such as changes in rainy days due to climate change.

Multinomial Distribution

- It is a generalized form of the binomial distribution.
- It is used when each trial has more than two possible outcomes (k).
- Here, the random variable is the no. of occurrences $(X_1, \dots, X_k)$ in a sequence of n independent experiments.
- Parameters are n and p_i $(i = 1, \dots, k)$, where p_i is the probability of occurrences of the ith outcome (X_i) in each experiment.
- The Probability Mass Function (PMF) is given by:

$$p(x_1, x_2, \dots, x_k) = \frac{n!}{x_1!\, x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

for $x_i = 0,1, \dots n$ and $\sum_{i=1}^{k} x_i = n$

Multinomial Distribution

- The Probability Mass Function (PMF) is also given by:

$$p(x_1, x_2, \dots, x_k) = \frac{\Gamma(\sum_i x_i + 1)}{\Pi_i \Gamma(x_i + 1)} \prod_{i=1}^{k} p_i^{x_i}$$

for $x_i = 0,1, \dots n$

- The CDF is expressed as:

$$F_X(x_i) = \sum_{X_i < x_i} p\ (x_1, x_2, \dots x_k)$$

- The mean and variance of the multinomial distribution are :

$E\ (X_i) = np_i$; $Var(X_i) = np_i(1 - p_i)$

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Multinomial Distribution

- For $n = 1$ and $k = 2$, multinomial distribution becomes Bernoulli Distribution.
- For $n > 1$ and $k = 2$, multinomial distribution becomes Binomial Distribution.
- For $n = 1$ and $k > 2$, it becomes the Categorical Distribution.

Applications:

- Used when hydrological variables are divided into more than two categories, such as very low, low, normal, high, and very high rainfall (multiple category event modelling).

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Conclusions

- Probability distributions describe uncertainty in real-world variables.
- PMF and CDF of various discrete probability distributions.
- Binomial and geometric-type distributions are useful for modelling occurrence and first-occurrence problems.
- Hypergeometric distribution is suitable for sampling without replacement from a finite population.
- Poisson distribution is useful for modelling rare events occurring within a fixed time or space interval.
- These distributions are widely applied in hydrology and civil engineering for analysing floods, rainfall events, drought years, cyclones, and infrastructure failures.





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 03: Probability Distributions: Discrete and Continuous distributions

LECTURE W3L2: Discrete Probability Distributions…contd.

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CONCEPTS COVERED

Discrete Probability Distributions: A Few example problems

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Example – 1

Let X is a discrete random variable and its probability distribution is given below:

X	0	1	2	3	4
$P(X = x)$	0.12	0.26	0.34	0.20	0.08

Determine:

a) whether the given probability distribution is valid.

b) what is the probability of $P(X = 2)$?

c) what is the probability of $P(X \geq 3)$?

d) what is the probability of $P(X \leq 1)$?

Solution

(a) Validity of PMF

For a valid probability distribution: $\sum_{all\ i} p_X(x_i) = 1$

Here, $\sum P(X = x) = 0.12 + 0.26 + 0.34 + 0.20 + 0.08 = 1$

Therefore, the given distribution is valid.

(b) Probability of $X = 2$, i.e., $P(X = 2) = 0.34$ (From the given table)

(c) $P(X \geq 3) = P(X = 3) + P(X = 4) = 0.20 + 0.08 = 0.28$

(d) $P(X \leq 1) = P(X = 0) + P(X = 1) = 0.12 + 0.26 = 0.38$

Example – 2

Number of rainy days in the last week of December, at a location is found to follow the following distribution.

$$p_X(x) = \begin{cases} C & \text{for } x = 0 \\ \frac{e^{-1}}{2x} & \text{for } x = 1, 2, \ldots, 7 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate the value of C for $p_X(x)$ to be a valid PMF and the probability of more than two rainy days in the last week of December at the location.

Solution

Let X be the number of rainy days in the last week of December.

The value of C can be evaluated as follows:

For the PMF to be valid, $\sum_{all\ i} p_X(x_i) = 1$

So, $C + \sum_{x=1}^{7} \frac{e^{-1}}{2x} = 1 \Rightarrow C + \frac{e^{-1}}{2\times1} + \frac{e^{-1}}{2\times2} + \frac{e^{-1}}{2\times3} + \frac{e^{-1}}{2\times4} + \frac{e^{-1}}{2\times5} + \frac{e^{-1}}{2\times6} + \frac{e^{-1}}{2\times7} = 1$

$\Rightarrow C + 0.1839 + 0.0919 + 0.0613 + 0.0459 + 0.0367 + 0.0306 + 0.0262 = 1$

$\Rightarrow C + 0.4765 = 1 \Rightarrow C = 1 - 0.4765 = 0.5235$

Hence, the PMF is

$$p_x(x) = \begin{cases} 0.5235 & for\ x = 0 \\ \frac{e^{-1}}{2x} & for\ x = 1, 2, \ldots, 7 \\ 0 & elsewhere \end{cases}$$

Solution

The probability of more than two rainy days denoted as $P(X > 2)$ can be evaluated as :

$$P(X > 2) = 1 - P(X \leq 2)$$
$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$
$$= 1 - \left[0.5235 + \frac{e^{-1}}{2} + \frac{e^{-1}}{2 \times 2}\right] = 1 - [0.5235 + 0.1839 + 0.0919]$$
$$= 0.201$$

Therefore, the probability of more than two rainy days in the last week of December at the location is 0.201.

Example - 3

During the last six decades (60 years), 22 instances of severe droughts were noticed in an ecosystem. The occurrence of each such instances is assumed to be temporally independent. Number of such droughts in a year (X) is assumed to follow a Poisson distribution.

a) What is the probability of noticing exactly 4 such droughts in next decade (10 years)?

b) What is the probability noticing more than 4 such droughts in next decade?

Solution

Poisson distribution:

$$p_X(x) = \lambda^x \frac{e^{-\lambda}}{x!} \qquad for \qquad x \geq 0$$

where λ is the expected number of droughts over a period.

It is given that, the 22 droughts in 60 years.

Therefore, $\lambda = 22/60 = 0.3667$

(a) Probability of exactly 4 droughts in next 10 years

For 10 years: $\lambda = 0.3667 \times 10 = 3.667$

Now, $P(X = 4) = \lambda^x \frac{e^{-\lambda}}{x!} = (3.667)^4 \frac{e^{-3.667}}{4!} = \frac{180.82 \times 0.0255}{24} = 0.192$

So, the probability of noticing exactly 4 severe droughts in that ecosystem in next decade is 0.192.

Solution

(b) Probability of noticing more than 4 droughts, we may calculate the CDF, which is as follows:

The CDF is expressed as:

$$F_X(x; p) = \sum_{i=0}^{x} p_X(i; p) = \sum_{i=0}^{x} \lambda^x \frac{e^{-\lambda}}{x!}$$

Putting, $x = 4$ will yield the probability of getting not more than 4 droughts as per the definition of CDF.

Thus, the probability of getting more than 4 droughts is as follows:

$$P(X > 4) = 1 - P(X \leq 4) = 1 - F_X(4) = 1 - \sum_{i=0}^{4} \lambda^i \frac{e^{-\lambda}}{i!}$$

Solution

$$P(X > 4) = 1 - \left((3.667)^0 \frac{e^{-3.667}}{0!} + (3.667)^1 \frac{e^{-3.667}}{1!} + (3.667)^2 \frac{e^{-3.667}}{2!} + (3.667)^3 \frac{e^{-3.667}}{3!} + (3.667)^4 \frac{e^{-3.667}}{4!} \right)$$

$$= 1 - 0.694 = 0.306$$

So, the probability of noticing more than 4 severe droughts in that ecosystem in next decade is 0.306.

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Example – 4

An ML based classification model is actively filtering incoming traffic for a corporate email server. Based on historical back testing, 4 out of 5 phishing emails are correctly flagged by the model. During a targeted attack, 5 distinct phishing emails hit the server simultaneously. Let X be the number of phishing emails successfully caught by the ML classifier, determine:

a) the probability that the ML classifier correctly flags exactly 4 of the phishing emails?

b) the probability that the model successfully catches all 5 phishing emails?

c) the probability that the classifier catches at most 4 of the phishing emails (meaning at least one bypasses the filter?

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Solution

Here, the random variable is X = number of phishing emails successfully caught by the ML classifier. X can take the values from 0 to 5.

Probability that a randomly selected phishing email is correctly flagged, $p = \frac{4}{5} = 0.80$.

Probability that a randomly selected phishing email bypasses the filter,

$$q = 1 - p = 1 - 0.8 = 0.20$$

No. of phishing emails hitting the server during the attack, $n = 5$

Binomial distribution:

$$p_X(x; n, p) = {}^{n}C_x \, p^x \, (1 - p)^{n-x}, \qquad x = 0,1,2, \ldots, n$$

Solution

(a) Probability that exactly 4 phishing emails are caught

Using the binomial distribution,

$$P(X = 4) = {}^{n}C_x \, p^x \, (q)^{n-x} = {}^{5}C_4 \, 0.8^4 \, (0.2)^{5-4} = \frac{5!}{4!\,(5-4)!} \times 0.4096 \times 0.2$$

$$= 5 \times 0.4096 \times 0.2 = 0.41$$

Therefore, the probability that exactly 4 phishing emails are caught is 0.41.

(b) Probability that all 5 phishing emails are caught

$$P(X = 5) = {}^{5}C_5 \, 0.8^5 \, (0.2)^{5-5} = \frac{5!}{5!\,(5-5)!} \times 0.32768 \times 1$$

$$= 1 \times 0.32768 \times 1 = 0.328$$

Therefore, the probability that all 5 phishing emails are correctly flagged is approximately 0.328.

Solution

(c) Probability that at most 4 phishing emails are caught

$$P(X \leq 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$
$$= 1 - P(X = 5) = 1 - 0.328 = 0.672$$

Thus, the probability that at most 4 phishing emails are caught (meaning at least one bypasses the filter) is approximately 0.672.

Example – 5

An ML engineer is using a Random Search algorithm to tune the hyperparameters of a deep neural network. The goal is to find a configuration that achieves a validation accuracy above a strict baseline. Based on previous experiments on this dataset, the probability that any randomly sampled hyperparameter configuration meets this baseline is 0.25. Let X be the number of hyperparameter configurations trained and evaluated until the first successful configuration is found. Determine:

a) What is the probability that the algorithm finds the first successful hyperparameter configuration on exactly the 3rd evaluation?

b) What is the probability that the very first configuration evaluated immediately meets the target accuracy?

c) What is the probability that the engineer finds a successful configuration within the first 4 evaluations?

Solution

Here, the random variable is,

X = number of hyperparameter configurations evaluated until the first successful configuration is found.

Probability that a randomly sampled hyperparameter configuration successfully meets the baseline, $p = 0.25$.

Geometric Distribution:

$$p_X(x; p) = p(1-p)^{x-1} \qquad for\ x = 1,2, \dots, n$$

a) Probability that the first successful configuration occurs on exactly the 3rd evaluation

Using the geometric distribution,

$$P(X = 3) = p(1-p)^{x-1} = (0.25)(1-0.25)^{3-1} = 0.25 \times 0.75^2$$
$$= 0.25 \times 0.5625 = 0.141$$

Therefore, the probability that the algorithm finds the first successful configuration on exactly the 3rd evaluation is approximately 0.141.

Solution

(b) Probability that the first successful configuration occurs on the 1st evaluation

$$P(X = 1) = (0.25)(1-0.25)^{1-1} = 0.25 \times 0.75^0 = 0.25 \times 1 = 0.25$$

Therefore, the probability that the very first configuration evaluated immediately meets the target accuracy is 0.25.

(c) Probability that the first successful configuration occurs within the first 4 evaluations

$$P(X \leq 4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$
$$= (0.25)(1-0.25)^{1-1} + (0.25)(1-0.25)^{2-1} +$$
$$(0.25)(1-0.25)^{3-1} + (0.25)(1-0.25)^{4-1}$$

Solution

$$P(X \leq 4) = 0.25 \times [0.75^0 + 0.75^1 + 0.75^2 + 0.75^3]$$
$$= 0.684$$

Thus, the probability that the engineer finds a successful configuration within the first 4 evaluations is approximately 0.684.

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MODULE 03: Probability Distributions: Discrete and Continuous distributions

LECTURE W3L3: Continuous Probability Distributions –Part I

BY

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CONCEPTS COVERED

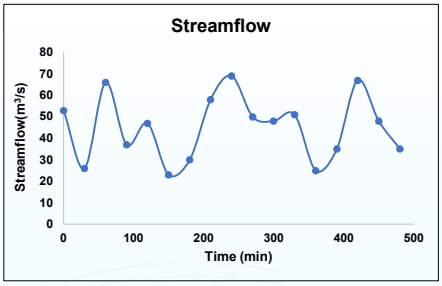
Continuous Probability Distributions

- **Uniform Distribution**
- **Exponential Distribution**
- **Normal and Log-Normal Distribution**
- **Gamma Distribution**
- **Pearson and Log Pearson Distribution**
- **Extreme Value Distribution**

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Continuous Random Variable

- If a random variable can take any possible real value from the range of real numbers, its probability distribution is called a continuous probability distribution.
- Mathematically, if X is a continuous univariate random variable, then it may take values such as: $X = 0.5, 1.25, 2.78, 10.6, \ldots$or any value within an interval $[a, b]$ over which it is defined.



Examples: Streamflow, temperature, speed, relative humidity, weights and biases in Neural Networks, etc.

- Support of these variables may be unbounded (e.g., temperature), bounded at one side (e.g., streamflow, speed) or both sides (e.g., relative humidity).

Continuous Probability Distribution

- A continuous probability distribution describes the probability behavior of a continuous random variable.
- In continuous probability distributions, probabilities are represented using a probability density function (pdf).
- The probability at a single exact point is always zero.
- Probabilities are calculated over an interval or range of a random variable using the area under the PDF curve.
- Continuous probability distributions are widely used in many fields, including machine learning applications in earth sciences

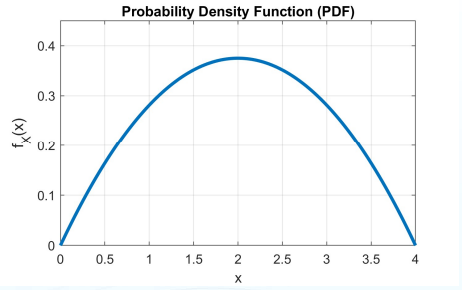
Probability Density Function (PDF)

- Let us consider X to be a continuous univariate random variable and x denotes a specific value of random variable.
- For any function to be a valid pdf, it has to satisfy two conditions as follows:

(i) $f_X(x) \geq 0 \quad for\ all\ \ x$

(ii) $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$

- $f_X(x)$ does not directly provide the value of probability, rather it is probability density.
- Integration over any range of x provides the probability of X being within that range.



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Cumulative Distribution Function (CDF)

- The CDF $(F_X(x))$ represents the probability that X is less than or equal to a specific value of x. This can be represented as

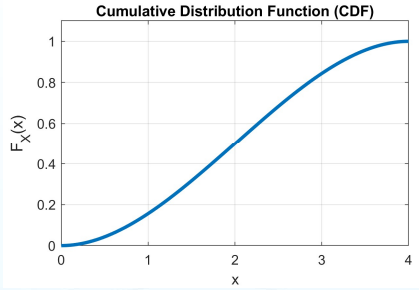
$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x)\, dx$$

- To obtain pdf from CDF, the CDF has to be differentiated with respect to x as follows:

$$\frac{d}{dx} F_X(x) = f_X(x)$$

- The probability that X lies between [a, b] is given by the following equation

$$P(a \leq X \leq b) = \int_{a}^{b} f_X(x)\, dx = F_X(b) - F_X(a)$$

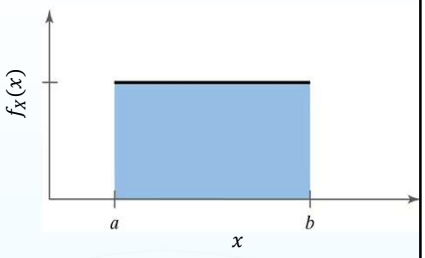


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Uniform Distribution

- In this continuous distribution, any value within a fixed interval has an equal probability of occurrence or equally likely.
- The random variable X is defined over a finite range(support), usually from a lower limit a to an upper limit b.
- Since all values are equally likely, the probability density remains constant throughout the interval.
- The random variable is X that is equi-probable over any subinterval of the same length within its support.
- The probability density function (PDF) of uniform distribution is:

$$f_X(x) = \frac{1}{b-a} \quad \text{where}, a \le x \le b$$



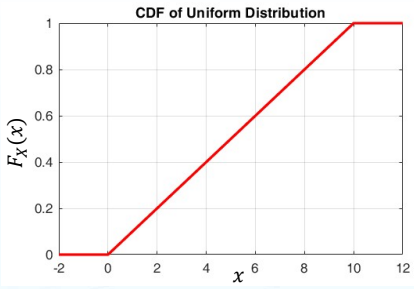
Uniform Distribution

- The Cumulative Distribution Function (CDF) is expressed as:

$$F_X(x) = \frac{x-a}{b-a} \quad \text{where}, a \le x \le b$$

- The mean, variance and coefficient of skewness of the uniform distribution are :

$$E(X) = \frac{a+b}{2} \qquad Var(X) = \frac{(b-a)^2}{12} \qquad \gamma = 0$$



Applications:

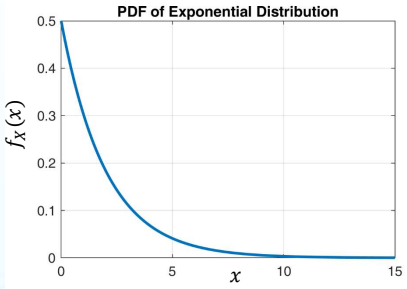
- Useful to generate random numbers (commonly, uniformly distributed over 0 to 1) for model initiation.
- In statistical analysis, p-value (to assess significance of statistical test) is uniformly distributed between 0 to 1, if the null hypothesis is true and distribution of the test statistic is continuous.

Exponential Distribution

- It is a continuous probability distribution, commonly used to model the time gap between two successive events.
- It is an asymmetric distribution.
- The random variable X represents the waiting time or inter-arrival time between two consecutive events.
- The probability density is higher near smaller values of x, meaning shorter waiting times are more likely than longer waiting times.
- The probability density function (pdf) is given by:

$$f_X(x) = \lambda e^{-\lambda}, \quad x \geq 0, \lambda > 0$$

where, λ is rate parameter, which is the average inter-arrival time (or space) between two successive events.



Exponential Distribution

- The CDF is expressed as:

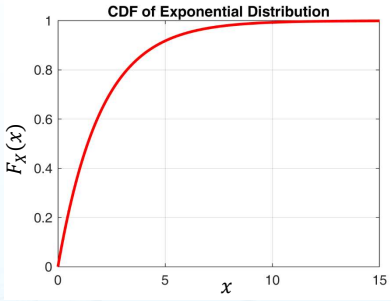
$$F_X(x) = 1 - e^{-\lambda}, \quad \text{for x} > 0$$

- The mean, variance and coefficient of skewness of the exponential distribution are :

$$E(X) = \frac{1}{\lambda} \quad Var(X) = \frac{1}{\lambda^2} \quad \gamma = 2$$

Applications:

- Useful for modelling inter-arrival time of any natural events like rainfall events, like, cyclone, earthquakes, etc. (temporal).
- Can be used to model spatial variation of a variable across a radial direction like variation in rainfall intensity from the storm center towards a particular radial direction, etc.(spatial).



Example – 1

A large deep learning model is retrained daily on new data. The time it takes for the model to reach an adequate, acceptable accuracy threshold varies due to various reason including data noise, and initial random weight initializations.

Let, X = the time required for the model to reach adequate accuracy, in hours.

Assume that X is a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} Cx(10-x), & 0 < x < 10 \\ 0, & otherwise \end{cases}$$

where C is a constant.

a) Find the value of C so that $f(x)$ is a valid PDF.

b) Find the probability that the time required for the model to reach adequate accuracy is between 3 hours and 7 hours.

c) Find the probability that the time required for the model to reach adequate accuracy exceeds 6 hours.

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Solution

a) Finding the value of C

For $f(x)$ to be a valid PDF: $\int_0^{10} f(x) \cdot dx = 1 \Rightarrow \int_0^{10} Cx(10-x).dx = 1$

$$\Rightarrow C\left[\frac{10^2}{2} - \frac{x^3}{3}\right]_0^{10} = 1 \Rightarrow C\left[\frac{10(10^2)}{2} - \frac{10^3}{3}\right] = 1 \Rightarrow C = \frac{3}{500} = 0.006$$

So, the valid PDF is $f(x) = 0.006x(10-x), 0 < x < 10$.

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b) Probability that the time required for the model to reach adequate accuracy is between 3 hours and 7 hours

$$P(3 < X < 7) = \int_3^7 0.006x(10-x).dx = 0.006\left[\frac{10x^2}{2} - \frac{x^3}{3}\right]_3^7$$

Solution

$$P(3 < X < 7) = 0.006 * (\frac{392}{3} - 36) = \frac{284}{500} = 0.568$$

So, the probability that the time required for the model to reach adequate accuracy is between 3 hours and 7 hours is 0.568.

c) Probability that the time required for the model to reach adequate accuracy exceeds 6 hours

$$P(X > 6) = \int_6^{10} 0.006x(10 - x).\,dx = 0.006 \left[\frac{10x^2}{2} - \frac{x^3}{3}\right]_6^{10}$$

$$= 0.006 * (\frac{500}{3} - 108) = \frac{176}{500} = 0.352$$

Therefore, the probability that the time required for the model to reach adequate accuracy exceeds 6 hours is 0.352.

Example – 2

In the same previous example, let X, i.e., the time taken for the model to reach adequate accuracy, follows an exponential distribution with mean time of 5 months.

(a) Write the PDF and CDF of X.

(b) Find the probability that the model to reach adequate accuracy within 3 hours.

(c) Find the probability that the model to reach adequate accuracy after 6 hours.

(d) Find the probability that the model to reach adequate accuracy between 2 hours and 7 hours.

Solution

(a) PDF and CDF

For an exponential distribution,

$$f(x) = \lambda e^{-\lambda}\ , x \geq 0$$

Here, $\lambda = \frac{1}{\text{Mean}} = \frac{1}{5} = 0.2$

Therefore, PDF: $f(x) = 0.2e^{-0.2x}, x \geq 0$

The CDF is:

$$F(x) = P(X \leq x) = \int_0^{\infty} 0.2e^{-0.2x}.\mathrm{d}x = 1 - e^{-0.2x}$$

Therefore,

$$F(x) = 1 - e^{-0.2x}, x \geq 0$$

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Solution

(b) Probability that the model to reach adequate accuracy within 3 hours

$P(X < 3) = F(3) = 1 - e^{-0.2x} = 1 - e^{-0.2(3)} = 1 - e^{-0.6} = 1 - 0.548 = 0.451$

So, the probability that the model to reach adequate accuracy within 3 hours is 0.451.

(c) Probability that that the model to reach adequate accuracy after 6 hours

$P(X > 6) = 1 - F(6) = 1 - (1 - e^{-0.2(6)}) = e^{-1.2} = 0.301$

Therefore, the probability that that the model to reach adequate accuracy after 6 hours is 0.301.

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Solution

(d) Probability that the model to reach adequate accuracy between 2 hours and 7 hours

$$P(2 < X < 7) = F(7) - F(2) = \left(1 - e^{-0.2(7)}\right) - \left(1 - e^{-0.2(2)}\right)$$
$$= e^{-0.4} - e^{-1.4} = 0.6703 - 0.2466 = 0.424$$

So, the probability that the model to reach adequate accuracy between 2 hours and 7 hours is 0.424.

CONCLUSIONS

- Continuous probability distributions describe uncertainty in variables that can take any value within a range.
- Uniform distribution is useful when all values within a fixed interval are equally likely.
- Exponential distribution is suitable for modelling waiting time or inter-arrival time between successive events.
- Other distributions will be discussed in the next lecture.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 03: Probability Distributions: Discrete and Continuous distributions

LECTURE W3L4: Continuous Probability Distributions –Part II

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Normal Distribution

- It is also known as the Gaussian distribution or bell-shaped distribution. This continuous distribution is symmetric about its mean.
- The random variable X can take any value from $-\infty \; to \; +\infty$. The values close to the mean have higher probability density.
- The shape of the normal distribution depends on two parameters: $\mu = mean, \sigma^2 = variance$
- The Probability Density Function (PDF) is given by:

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

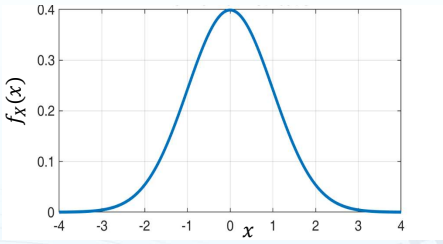
- The CDF is expressed as:

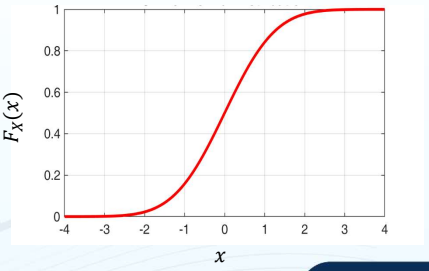
$$F_X(x; \mu, \sigma^2) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx, \; -\infty < x < \infty$$

- The coefficient of skewness of the normal distribution is zero.

Normal Distribution

When $\mu = 0, \; \sigma^2 = 1$, it is called standard normal distribution.





Applications:

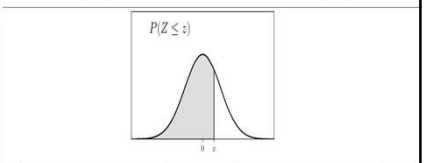
- Many statistical methods are developed under the assumption that sample data follows normal distribution.
- In natural systems, many variables may be found to follow normal distribution, e.g., temperature, wind velocity etc.
- Error series from any model is generally considered to follow standard normal distribution.

Normal Distribution

- The normal probability distribution cannot be integrated into closed form between two limits, say a to b.
- Therefore, the probabilities related to normal distribution are computed numerically.
- Those values are used to prepare standard normal distribution table (sample given in figure).
- Any normally distributed variable can be converted into a standard normal variable using: $z = \frac{X-\mu}{\sigma}$
- To find the probability that a random variable having the standard normal distribution will take on a value between a and b, the following equation can be used-

$$P\ (a\ <\ Z\ \leq\ b) = F\ (b) -\ F(a)$$

where $F(b)$ and $F(a)$ values are obtained from table.



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879

Example - 1

Considering the time required (X) (in minutes) to train a single epoch of a deep neural network follows a normal distribution with a mean of 10 minutes and a standard deviation of 5 minutes, determine:

a) What is the probability that a given epoch takes between 15 minutes and 24 minutes to train?

b) What is the probability that an epoch takes greater than 5 minutes to train?

c) What is the probability that an epoch finishes training in less than 20 minutes?

Solution

We can transform the RV X to a standard normal distribution, Z and then use standard normal tables.

Given: $X \sim N(\mu, \sigma)$ with $\mu = 10$ min and $\sigma = 5$ min

(a) Probability that a given epoch takes between 15 min and 24 min to train

$x = 15$ transforms to

$$z = \frac{x - \mu}{\sigma} = \frac{15 - 10}{5} = \frac{5}{5} = 1$$

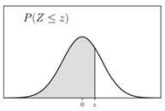
Similarly, $x = 24$ transforms to

$$z = \frac{x-\mu}{\sigma} = \frac{24-1}{5} = \frac{14}{5} = 2.8$$

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Solution

Standard Normal Distribution Table



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Solution

Probability that a given epoch takes between 15 min and 24 min to train

$$P\ (15\ \leq\ X\ \leq 24) = P\left(\frac{15-}{5} \leq \frac{x-\mu}{\sigma} \leq \frac{24-10}{5}\right) =\ P(1 \leq\ Z\ \leq 2.8) = F_Z(2.8) - F_Z(1)$$

From standard normal distribution table, it can be seen that

$$F_Z(1) =\ 0.841 \text{ and } F_Z(2.8)\ =\ 0.997.$$

Thus, the probability is,

$$P\ (15\ \leq\ X\ \leq 24) =\ P(1 \leq\ Z\ \leq 2.8) = F_Z(2.8) - F_Z(1)$$

$$=\ 0.997 - 0.841\ =\ 0.156$$

So, the probability that a given epoch takes between 15 minutes and 24 minutes to train is 0.156.

Solution

(b) Probability that an epoch takes greater than 5 min to train

$x\ =\ 15$ transforms to $z\ =\ \frac{x-\mu}{\sigma} = \frac{5-10}{5} = \frac{-5}{5} = -1$

Thus, the desired probability is:

$$P(X > 5) = 1 - P(X \leq 5) = 1 - P\left(\frac{x-\mu}{\sigma} \leq \frac{5-1}{5}\right) = 1 - P(Z \leq -1) = 1 - F_Z(-1)$$

$$= 1 - 0.159 = 0.841$$

So, the probability that an epoch takes greater than 5 minutes to train is 0.841.

(c) Probability that an epoch finishes training in less than 20 minutes

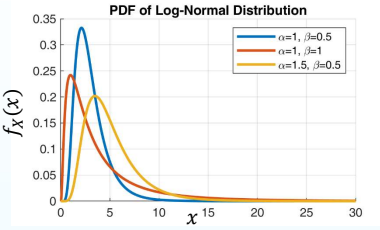
$x\ = 20$ transforms to $z\ =\ \frac{x-\mu}{\sigma} = \frac{20-10}{5} = \frac{10}{5} = 2$

Thus, the desired probability is:

$$P(X < 20)\ =\ P(Z \leq 2)\ = F_Z(2) = 0.977$$

Therefore, the probability that an epoch finishes training in less than 20 minutes is 0.977.

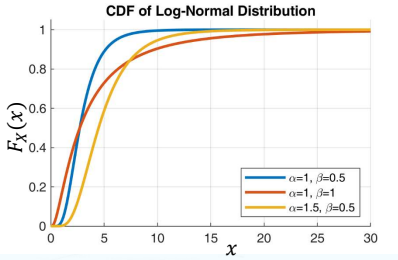
Log-Normal Distribution



- Log-normal distribution is a continuous probability distribution used for variables that take only positive values.
- A random variable X is said to follow a log-normal distribution if its logarithmic transformation $(Y = \ln(X))$ follows a normal distribution.
- It is positively skewed, which means the smaller values are more likely than the larger values.
- The two main parameters are: α = mean of $\ln(X)$ and β = standard deviation of $\ln(X)$.
- The Probability Density Function (PDF) is given by:

$$f_X(x) = \begin{cases} \frac{1}{x\sqrt{2\pi\beta^2}} e^{-\frac{(\ln x - \alpha)^2}{2\beta^2}} & for\ x > 0;\ \alpha, \beta > 0 \\ 0 & elsewhere \end{cases}$$

Log-Normal Distribution



- The CDF is expressed as:

$$F_X(x) = \int_0^x f_X(x)dx \qquad for\ x > 0; \alpha, \beta > 0$$

- The mean, variance and coefficient of skewness of the Log – Normal distribution are :

$$\mu = e^{(\frac{\alpha+\beta^2}{2})} \qquad \sigma^2 = (e^{\beta^2} - 1)e^{(2\alpha+\beta^2)}$$

$$\gamma = (e^{\beta^2} + 2)\sqrt{e^{\beta^2} - 1}$$

- Since the logarithm of the variable follows a normal distribution, probability calculations are usually done after converting X into $\ln(X)$.

Applications:

It is mostly applicable for hydrologic variables like monthly rainfall depth, river discharge volumes, extreme values, etc.

Example – 2

The inference latency (X) (time taken to process a single request) for a deployed Random Forest model in a production environment is found to have a mean of 130 milliseconds (ms) and a standard deviation of 30 ms. Considering the inference latency is strictly positive and typically right-skewed, it is modeled using a lognormal distribution. Evaluate the following:

a) Probability of the inference latency being greater than 180 ms (which would trigger a system timeout).

b) Probability of the inference latency lying in between 120 ms and 150 ms.

Solution

Given $\bar{x} = 130$ and $S_X = 30$

As the inference latency follows log-normal distribution, the parameters ($\bar{y}$ and S_Y) can be evaluated from the sample statistics ($\bar{x}$ and S_X) as follows:

$$C_v = \frac{S_X}{\bar{x}} = \frac{30}{130} = 0.231$$

$$\bar{y} = \frac{1}{2}\ln\left[\frac{\bar{x}^2}{{C_v}^2 + 1}\right] = \frac{1}{2}\ln\left[\frac{130^2}{0.231^2 + 1}\right] = 4.841$$

$$S_Y = \sqrt{\ln\left({C_v}^2 + 1\right)} = \sqrt{\ln(0.231^2 + 1)} = 0.228$$

Solution

(a) Probability of the inference latency being greater than 180 ms (which would trigger a system timeout)

For $x = 180$, the reduced variate is

$$Z = \frac{y - \bar{y}}{S_Y} = \frac{\ln x - \bar{y}}{S_Y} = \frac{\ln 180 - 4.841}{0.228} = 1.544$$

Thus, the desired probability is:

$$P(Y > \ln 180) = 1 - P(Y \leq 180) = 1 - P(Z < 1.544)$$

$$= 1 - 0.939 = 0.061$$

Therefore, the probability of the inference latency being greater than 180 ms (which would trigger a system timeout) is approximately 0.06.

Solution

(b) Probability of the inference latency lying in between 120 ms and 150 ms

For $x_1 = 120$ and $x_2 = 150$, the corresponding reduced variates are

$$Z_1 = \frac{y_1 - \bar{y}}{S_Y} = \frac{\ln x_1 - \bar{y}}{S_Y} = \frac{\ln 120 - 4.841}{0.228} = -0.235$$

$$Z_2 = \frac{y_2 - \bar{y}}{S_Y} = \frac{\ln x_2 - \bar{y}}{S_Y} = \frac{\ln 150 - 4.841}{0.228} = 0.744$$

Thus, the desired probability is:

$$P(\ln 120 < Y < \ln 150) = P(-0.235 < Z < 0.744)$$

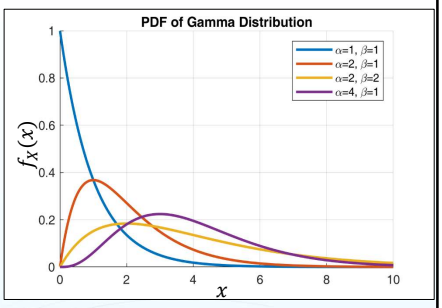
$$= F_Z(0.744) - F_Z(-0.235) = 0.7704 - 0.4052 = 0.365$$

Therefore, the probability of the inference latency lying in between 120 ms and 150 ms is approximately 0.365.

Gamma Distribution

- It is a continuous probability distribution that is positively skewed over the positive side of the real line. Thus, the random variable (X) is continuous and positively skewed that takes non-negative values only.
- It can be treated as the sum of exponentially distributed random variables each with the same parameter.
- The parameter α can be interpreted as the number of random variables following exponential distribution and β is the inverse of parameter of the exponential distributions.
- The probability density function of gamma distribution is:

$$f_X(x) = \begin{cases} \dfrac{x^{\alpha-1}e^{-x/\beta}}{\beta^{\alpha}\Gamma(\alpha)} & for\ x \geq 0; \alpha, \beta > 0 \\ 0\ elsewhere \end{cases}$$



Gamma Distribution

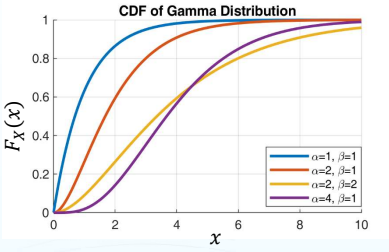
- $\Gamma(\alpha)$ is the value of the gamma function defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1}e^{-x}dx = (\alpha-1)\Gamma(\alpha-1)$$

- Exponential distribution is a special case of gamma distribution with $\alpha = 1$.
- The CDF of gamma distribution is expressed as :

$$F_X(x) = \int_0^x f_X(x)dx \quad for\ x \geq 0; \alpha, \beta > 0$$

- The mean, variance and coefficient of skewness of the gamma distribution are : $\mu = \alpha\beta; \quad \sigma^2 = \alpha\beta^2; \quad \gamma = \frac{2}{\sqrt{\alpha}}$



Applications:

- Useful for modelling of variables which are always positive and lower bounded by zero (rainfall, river discharge, etc.).

Example – 3

An online fraud detection system utilizes two servers. Based on the past performance statistics, each server may fail once in every 100 days of operation on an average. The time gaps between two successive failures of each server may be assumed to follow an exponential distribution. What is the probability that the entire two-server system fails before 80 days of operation?

Solution

Since the time gaps between two successive failures of each server follows exponential distribution, the time span between two successive failures for the entire two-server system should follow a gamma distribution.

Solution

Given that, the mean inter-arrival time between two successive failures is 100 days.

Thus,

$$\bar{x} = \frac{1}{\lambda} = 100 \qquad \Rightarrow \lambda = \frac{1}{100}$$

The failure time of each server may be assumed to follow an exponential distribution with: $\lambda = \frac{1}{100}$.

Since the online fraud detection system has two servers, the system fails only after both servers fail.

Therefore, the total time until system failure, that follows a gamma distribution with: $\alpha = 2;\ \beta = \frac{1}{\lambda} = 100$.

Let Y denote the time elapsed until the entire two-server system failure, that is, the failure of both the servers.

Solution

The probability density function of Y is:

$$f_Y(y) = \frac{y^{\alpha-1}e^{-y/\beta}}{\beta^{\alpha}\Gamma(\alpha)} = \frac{y^{2-1}e^{-y/100}}{100^2\Gamma(2)} = \frac{ye^{-y/100}}{10000 \times 1} = \frac{ye^{-y/100}}{10000} \Rightarrow f_Y(y) = \frac{ye^{-y/100}}{10000}$$

Therefore, the probability that the two-server system fails before 80 days is:

$$P(Y \leq 80) = \int_0^{80} \frac{ye^{-y/100}}{10000} dy$$

Solving using integration by parts:

$$P(Y \leq 80) = \frac{1}{10000}\left[\frac{ye^{-y/100}}{(-1/100)} - \frac{e^{-y/100}}{(-1/100)^2}\right]_0^{80}$$

Solution

Now, substituting the limits:

$$P(Y \leq 80) = \frac{1}{10000}\left[\frac{80e^{-8/100}}{(-1/100)} - \frac{e^{-8/100}}{(-1/100)^2} - \left(\frac{0e^{-0/100}}{(-1/100)} - \frac{e^{-0/100}}{(-1/100)^2}\right)\right]$$

$$= \frac{1}{10000}[-8000e^{-0.8} - 10000e^{-0.8} + 10000] = \frac{1}{10000}[10000 - 18000e^{-0.8}]$$

$$= 1 - 1.8e^{-0.8} = 1 - 1.8 \times 0.4493 = 1 - 0.808 = 0.192$$

Therefore, the probability that the online fraud detection system fails before completing 80 days of operation is 0.192.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 03: Probability Distributions: Discrete and Continuous distributions

LECTURE W3L5: Continuous Probability Distributions – Part III

BY

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CONCEPTS COVERED

Continuous Probability Distributions

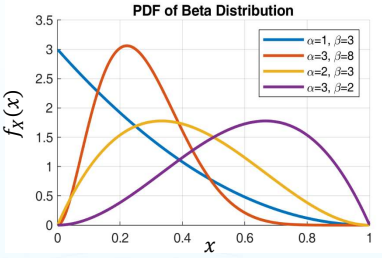
- **Uniform Distribution**
- **Exponential Distribution**
- **Normal and Log-Normal Distribution**
- **Gamma Distribution**
- **Beta Distribution**
- **Pearson and Log Pearson Distribution**
- **Extreme Value Distribution**

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Beta Distribution

- It is a continuous probability distribution that represents outcomes for percentages or proportions over an interval, parameterized by two shape parameters.
- The random variable (X) is continuous and generally defined within the interval $[0, 1]$.
- The parameters α and β are shape parameters of this distribution.
- The probability density function of beta distribution is:

$$f_X(x) = \begin{cases} \dfrac{\Gamma(\alpha+\beta)x^{\alpha-1}(1-x)^{\beta-1}}{\Gamma(\beta).\Gamma(\alpha)} & \text{for } 0 < x < 1; \alpha, \beta > 0 \\ 0 & elsewhere \end{cases}$$



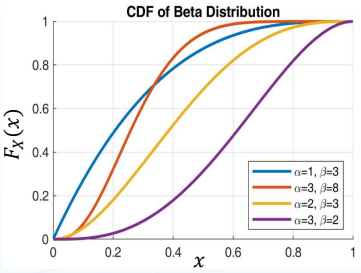
Beta Distribution

- The beta distribution can also be transformed to any interval (a, b).
- The shape parameters of the distribution vary with the nature of the distribution.
- The CDF of beta distribution is expressed as :

$$F_X(x) = \int_0^x f_X(x)dx \qquad for\ 0 < x < 1; \alpha, \beta > 0$$

- The mean, variance and coefficient of skewness of the beta distribution are :

$$E(X) = \frac{\alpha}{\alpha+\beta};\ Var(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2};\quad \gamma = \frac{2(\beta-\alpha)\sqrt{\alpha+\beta+1}}{\sqrt{\alpha\beta}(\alpha+\beta+2)}$$



Applications:

- Useful for modelling of variables limited to intervals of finite length, for example, volumetric soil moisture content that varies between 0 and 1.

Pearson Type–III Distribution

- It is one of the seven different types of Pearson distribution. These are continuous probability distributions.
- The **Pearson Type–III Distribution** is skewed and the mode of the data is at zero.
- The parameters λ and β are scale and shape parameters of this distribution.
- The probability density function (PDF) is:

$$f_X(x) = \frac{\lambda^\beta (x-\varepsilon)^{\beta-1} e^{-\lambda(x-\varepsilon)}}{\Gamma(\beta)} \quad for\ x \geq \varepsilon$$

Applications:

- Mainly used for frequency analysis, for example, annual maximum flood peaks.

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Log-Pearson Type – III Distribution

- It is also a continuous probability distribution
- If a random variable $Y = \log(X)$ follows Pearson Type – III distribution, then the random variable X follows the log-Pearson Type – III distribution.
- The probability density function is:

$$f_X(x) = \frac{\lambda^\beta (y-\varepsilon)^{\beta-1} e^{-\lambda(y-\varepsilon)}}{\Gamma(\beta)} \quad for\ y \geq \varepsilon, \text{ where } y = \log(x)$$

Applications:

- Mainly used for frequency analysis and modelling of highly positively skewed data, because, log transformation reduces the skewness and can even change a positively skewed data to a negatively skewed one .

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Extreme Value Distribution

- It is a continuous probability distribution used for the analysis of extreme values.
- The extreme values (Y) from a set of random variables (X) can also be assumed to be random.
- The probability distribution of these extreme values depends on the size of the sample (n) and the distribution from which the sample is drawn.
- For a random sample of X having size n, let Y be the largest of the sample values.
- Now, $P(Y \leq y) = F_Y(y)$ and $P(X_i \leq x) = F_{X_i}(x)$
- Hence, $F_Y(y) = P(Y \leq y) = P(\text{all possible values of x} \leq y)$
- If the $x's$ are independently and identically distributed, we have

$$F_Y(y) = [F_X(y)]^n$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = n[F_X(y)]^{n-1} f_X(y)$$

Extreme Value Distribution

- However, the parent distribution from which the extreme value is observed is not known and cannot be determined.
- In such cases, if the sample size is large, we can use certain general asymptotic results that depend on limited assumptions concerning the parent distribution of extreme values.
- Three types of asymptotic distributions have been developed based on different parent distribution:

(i) Type I — parent distribution unbounded in direction of the desired extreme and all the moments of the distribution exist.

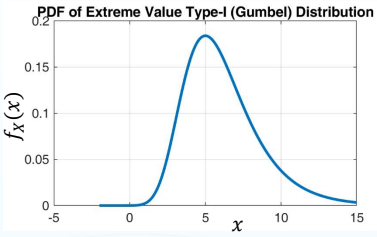
(ii) Type II — parent distribution unbounded in direction of the desired extreme and all the moments of the distribution do not exist.

(iii) Type III —parent distribution bounded in the direction of the desired extreme.

Extreme Value Type-I Distribution

- Also, known as Gumbel distribution, is a limiting probability distribution used to model the maximum or minimum values from a sample of independent, identically distributed random variables, as the size of the sample increases.
- The continuous random variable (X), which is the maximum/minimum of a number of samples of various distribution (e.g., normal or exponential).
- The parameters α and β are scale and location parameters, where $\beta - \alpha\ln(\ln 2)$ is the median of the distribution and β is the mode of the distribution.
- The probability density function of Gumbel distribution is:

$$f_X(x) = \frac{1}{\alpha}\exp\left[\mp\frac{x-\beta}{\alpha} - \exp\left(\mp\frac{x-\beta}{\alpha}\right)\right]$$



Extreme Value Type-I Distribution

where $-\infty < x < \infty;\ -\infty < \beta < \infty;\ \alpha > 0$.

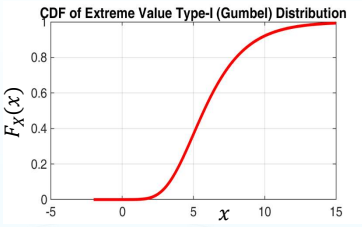
The negative sign implies maximum value, and the positive sign implies minimum value.

- The CDF is expressed as :

$$F_X(x) = \exp\left[\mp\exp\left(\mp\frac{x-\beta}{\alpha}\right)\right],\ \text{where } -\infty < x < \infty;\ -\infty < \beta < \infty;\ \alpha > 0$$

- The mean, variance and coefficient of skewness of the Gumbel distribution are :

$$E(X) = \beta \pm 0.5772\alpha;\ Var(X) = 1.645\alpha^2;\quad \gamma = \pm 1.1396$$



Applications:

- used to analyze variables such as monthly and annual maximum values of daily rainfall or river discharge volumes.
- used in the frequency analysis of floods.

Example – 1

In a certain stream, the annual maximum daily discharge follows Gumbel distribution with mean value of 18000 m^3/s and a standard deviation of 6000 m^3/s.

a) What is the probability that the annual maximum daily discharge will exceed 20000 m^3/s ?

b) What is the magnitude of annual maximum daily discharge with a return period of 130 years?

Solution

Given:

Random Variable : X = annual maximum daily discharge (in m^3/s)

Mean, E(X) = 18000 m^3/s

Standard Deviation = 6000 m^3/s

For Gumbel distribution, the parameters α and β are evaluated as follows:

$$Var(X) = 1.645\alpha^2 \Rightarrow 6000^2 = 1.645\alpha^2 \Rightarrow \alpha = \sqrt{\frac{6000^2}{1.645}} = 4678.1$$

Also, $E(X) = \beta + 0.5772\alpha$

$$\Rightarrow 18000 = \beta + 0.5772 \times 4678.1 \Rightarrow \beta = 18000 - 0.5772 \times 4678.1$$

$$\Rightarrow \beta = 15299.8$$

Solution

(a) Probability that the annual maximum daily discharge will exceed 20000 m³/s

$$P(X > 20000) = 1 - P(X \leq 20000) = \quad 1 - F_X(20000)$$

$$= 1 - \exp\left(-\exp\left(-\frac{(20000-15299.8)}{4678.1}\right)\right)$$

$$= \ 1 - 0.6934$$

$$= 0.3066$$

Therefore, the probability that the annual maximum daily discharge will exceed 20000 m³/s is approximately 0.307

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Solution

b) The magnitude of annual maximum daily discharge with a return period of 130 years

Let the magnitude with return period 130 years be x.

Then, the $P\ (X\ >\ x)\ =\ 1/130\ =\ 0.0077,$

Hence, $P\ (X\ > x) = 1 - P(X\ \leq x)$

$$\Rightarrow\ 1 - \exp\left(-\exp\left(-\frac{(x-15299.8)}{4678.1}\right)\right) = 0.0077$$

$$\Rightarrow 0.9923 = \exp(-\exp(-\frac{(x-1\quad .8)}{4678.1}))$$

$$\Rightarrow x = 38047.87\ m^3/s$$

Therefore, the magnitude of of annual maximum daily discharge with a return period of 130 years is approximately 38048 m³/s.

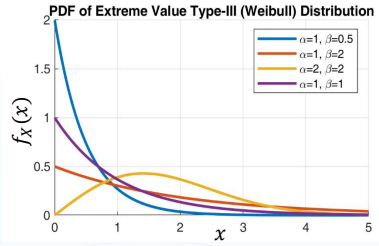
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Extreme Value Type – III Distribution

- Also, known as Weibull distribution, is utilized for the extremes in the direction towards which the parent distribution is limited.
- It is generally used for modelling minimum values in engineering.
- The random variable (X) is continuous and is the minimum of a sample from an asymmetric distribution and takes non-negative values.
- The parameters α and β are shape and scale parameters.
- The probability density function of Weibull distribution is:

$$f_X(x) = \alpha x^{\alpha-1}\beta^{-\alpha}\exp[-(x/\beta)^{\alpha}]$$

$$where\ x \geq 0;\ \alpha, \beta > 0$$



Extreme Value Type – III Distribution

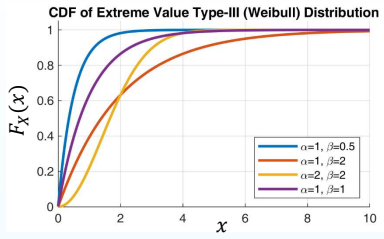
- The nature of the distribution varies with the change in the shape and scale parameters.
- The CDF is expressed as :

$$F_X(x) = 1 - \exp[-(x/\beta)^{\alpha}],\ \ where\ x \geq 0;\ \alpha, \beta > 0$$

- The mean, variance and coefficient of skewness of the Weibull distribution are : $E(X) = \beta\Gamma(1 + {}^{1}/_{\alpha})$;

$$Var(X) = \beta^2[\Gamma(1 + {}^{2}/_{\alpha}) - \Gamma^2(1 + {}^{1}/_{\alpha})];$$

$$\gamma = \frac{\Gamma(1 + {}^{3}/_{\alpha}) - 3\Gamma(1 + {}^{2}/_{\alpha})\Gamma(1 + {}^{1}/_{\alpha}) + 2\Gamma^3(1 + {}^{1}/_{\alpha})}{\left[\Gamma(1 + {}^{2}/_{\alpha}) - \Gamma^2(1 + {}^{1}/_{\alpha})\right]^{3/2}}$$



Applications:

- used most efficiently in hydrology for analysis of low flows in the rivers, as the low flows are naturally lower bounded by zero.

Example – 2

A data engineering team is monitoring a deployed classification model in production. Over time, changes in real-world data distributions (concept drift) cause the model's accuracy to degrade. The continuous random variable X, representing the time in months until the model's accuracy drops below the acceptable threshold (triggering mandatory retraining), is modeled using a Weibull distribution. Based on historical monitoring, the distribution has a shape parameter $\alpha = 1.5$ (indicating that the rate of drift accelerates over time) and a scale parameter $\beta = 8$ months.

a) What is the probability that the deployed model operates for more than 10 months without needing to be retrained?

b) What is the probability that concept drift triggers the retraining pipeline between 6 and 12 months of deployment?

Solution

Given:

$X =$ time in months until the model's accuracy drops below the acceptable threshold (triggering mandatory retraining)

$\alpha = 1.5; \quad \beta = 8\, months$

For Weibull distribution, the CDF is:

$F_X(x) = 1 - \exp[-(x/\beta)^{\alpha}], \quad where\ x \geq 0;\ \alpha, \beta > 0$

$\Rightarrow F_X(x) = 1 - \exp[-(x/8)^{1.5}]$

(a) Probability that the deployed model operates for more than 10 months without needing to be retrained

$$P(X > 10) = 1 - P(X \leq 10) = 1 - F_X(10)$$

Solution

$$P(X > 10) = 1 - (1 - exp[-(10/8)^{1.5}]) = \exp(-1.3975)) = 0.247$$

Therefore, the probability that the deployed model operates for more than 10 months without needing to be retrained is 0.247.

b) Probability that retraining occurs between 6 and 12 months

$$P(6 < X < 12) = F_X(12) - F_X(6) = 1 - \exp[-(12/8)^{1.5}] - (1 - \exp[-(6/8)^{1.5}])$$
$$= 1 - \exp(-(1.5)^{1.5}) - (1 - \exp(-(0.75)^{1.5}))$$
$$= 0.8407 - 0.4777 = 0.363$$

Therefore, the probability that retraining occurs between 6 and 12 months is 0.363.

